Melooha Engineering Challenge — Final Submission Report

Theme: 'Help more people discover Melooha, trust it, and come back to it.'

Deliverable	Details & Production URLs
Live Web Application	**https://astro-psi-olive.vercel.app** (Mobile-First SPA on Vercel)
Backend API Service	**https://astro-5qqa.onrender.com** (FastAPI Docker on Render)
GitHub Repository	**https://github.com/AkshatRabgotra/Astro** (Public repo, 74 tests passing)
Core Target Metric	**Day-7 (D7) Retention & Weekly Active Return Rate (WARR)**
Selected Feature	**Personalized Muhurta Advisor (Google Calendar Sync) + Daily Rewards Streak**

1. App Exploration & Problem Identification

User Drop-off Points in Melooha: Exploring the user experience highlights two key friction moments: (1) *Birth Time Uncertainty* during onboarding, and (2) the *Single-Question Drop-off*. Once a user reads their single free answer, a credit paywall appears. Because a birth chart feels like static information, users feel their questions have been answered and have **no daily reason to return tomorrow**.

Competitor Analysis & Market Gaps:

App	What They Do Well	Where They Fall Short	How Our Build Solves It
AstroTalk	Instant human astrologer chat.	Expensive per-minute timers; aggressive sales pushes for poojas and gems.	**Private AI**: Immediate readings with zero upsell pressure.
AstroSage	Exhaustive traditional chart tables.	Outdated 1990s UI cluttered with banner ads; complex jargon.	**Clean modern UI**: Translates planetary math into practical advice.
Astroyogi	Directory of verified consultants.	Requires appointment booking; high friction and slow turnaround.	**Instant answers**: Sub-second streaming readings 24/7.
Kundali GPT	Free-form natural language chat.	High rate of AI hallucinations; invents planetary positions.	**In-code verification**: C-engine math + AST code check.
Naksh	Minimalist modern aesthetic.	Static daily horoscope; lacks practical decision tools.	**Actionable utility**: Daily Muhurta with Google Calendar export.

2. The Idea: From a One-Time Reading to a Daily Decision Utility

Target User: Young professionals, students, and planners who want practical Vedic timing for day-to-day decisions (interviews, signing contracts, travel).
The Primary Metric: Day-7 (D7) Retention & Weekly Active Return Rate (WARR).
The Core Features Built:

- **Personalized Muhurta Advisor (1-Click Google Calendar Sync):** Computes daily 30-min astronomical windows (Choghadiya, Tithi, Nakshatra, and Tara Bala based on the user's Moon). Users click *Add to Google Calendar* to schedule auspicious times directly into their day.
- **Daily Rewards & Habit Streak:** Encourages daily app opens with consecutive check-in streaks (+1 free question credit, daily spiritual mantras), removing the feeling of an abrupt paywall.
- **Past Check & Dasha Twin:** Builds trust by validating the user's chart against known past life milestones before predicting future periods, and matches historical figures sharing the same planetary dasha.

3. System Architecture & Engineering Trade-offs

Layer	Components & Flow
Client (Frontend)	React 18 + Vite SPA · Tailwind CSS · Google Identity Services (One-Tap Login) · Deployed on **Vercel**
API Gateway	Vercel Reverse Proxy (/api/*) transparently routing requests to the Render backend without CORS friction.
Backend Service	FastAPI in Docker on **Render** · Python 3.11 · SSE Streaming Engine for instant sub-second text delivery.

Calculation & AI	**Swiss Ephemeris (pysweph C-Engine)** for exact planetary math (<15ms) · **Model Orchestrator** (Gemini 1.5 $\rightarrow$ Groq Llama-3.3 $\rightarrow$ Offline fallback) · **Python AST Guardrail** verifying facts before rendering.
Data Storage	**Neon Cloud Serverless PostgreSQL** · Idempotent question balance ledger · User profile and telemetry storage.

Key Engineering Decisions:
1. *In-Code Fact Verification*: Instead of relying on prompt instructions alone, true placements are calculated in C using Swiss Ephemeris, and generated text is parsed by an AST guardrail in Python.
2. *Multi-Provider Resiliency*: Model calls automatically cascade from Gemini 1.5 to Groq to an offline deterministic engine, ensuring the user never sees a raw error.
3. *Double-Entry Balance Ledger*: Prevents duplicate deductions or race conditions during rapid clicks.

4. Challenge Standout Features & Evaluation

• **Background Jobs & Calendar Sync:** Generates RFC-5545 calendar alerts and manages idempotent daily streak tracking.
• **Automated Evaluation Suite:** 74 unit and integration tests passing in CI (pytest -q in 4.6s) across calculations, guardrails, and API flows.
• **Living Design System & Bilingual Support:** Documented cosmic design tokens in the Admin Console; native English and Hindi support across all features.
• **Single Command Local Setup:** Run locally with .\run.ps1 (Windows) or make run (macOS/Linux).